

THE UNIVERSITY OF ZAMBIA
Department of Mathematics and Statistics
MAT1110: Foundation Mathematics and Statistics for Social Sciences
Tutorial Sheet 5 (2020/2021)

1. Find the quadrant that contains the terminal (end) side of θ if the given conditions are satisfied:

- | | | |
|---|---|---|
| (a) i. | iii. | v. |
| $\sin \theta > 0$ and $\tan \theta < 0$ | $\cos \theta > 0$ and $\tan \theta < 0$ | $\sin \theta < 0$ and $\cot \theta < 0$ |
| ii. | iv. | vi. |
| $\sin \theta < 0$ and $\cos \theta < 0$ | $\tan \theta > 0$ and $\sec \theta < 0$ | $\csc \theta > 0$ and $\sec \theta > 0$ |

(b) Without the use of a calculator, convert each of the following to radians:

- | | | |
|-----------------|------------------|-----------------|
| i. -150° | iii. 15° | v. 570° |
| ii. 225° | iv. -330° | vi. 135° |

(c) Without the use of a calculator, convert each of the following to degrees:

- | | | |
|-----------------------|-----------------------|------------------------|
| i. $\frac{4\pi}{3}$ | iii. $\frac{\pi}{12}$ | v. $\frac{7\pi}{6}$ |
| ii. $-\frac{3\pi}{4}$ | iv. $\frac{5\pi}{3}$ | vi. $-\frac{11\pi}{4}$ |

2. (a) Without using a calculator or table, find the exact values of each of the following:

- | | | |
|-----------------------|---------------------------|------------------------------|
| i. $\sin 225^\circ$ | iv. $\sin(-135^\circ)$ | vii. $\cot(-\frac{5\pi}{3})$ |
| ii. $\cos 225^\circ$ | v. $\tan 315^\circ$ | viii. $\sec \frac{7\pi}{6}$ |
| iii. $\cos 150^\circ$ | vi. $\cos \frac{7\pi}{3}$ | ix. $\csc(\frac{\pi}{12})$ |

(b) Simplify each of the following to a single trigonometric function or a constant:

- | | | |
|------------------------------|------------------------------------|---|
| i. $\sec x - \sin x \tan x$ | iii. $\cos x + \tan x \sin x$ | v. $\frac{\sec \theta - \cos \theta}{\tan \theta}$ |
| ii. $\csc x - \cos x \cot x$ | iv. $(\sin^2 x - 1)(\tan^2 x + 1)$ | vi. $\frac{\sec \theta + 1}{\tan \theta + \sin \theta}$ |

(c) Prove each of the following identities:

- | | |
|--|---|
| i. | v. |
| $\tan^2 x + 1 + \tan x \sec x = \frac{1 + \sin x}{\cos^2 x}$ | $\frac{(\tan x)(1 + \cot^2 x)}{1 + \tan^2 x} = \cot x$ |
| ii. | vi. |
| $\frac{\cos x + \sin x}{\cos x} = 1 + \tan x$ | $\frac{1 - \sin x}{1 + \sin x} = (\tan x - \sec x)^2$ |
| iii. | vii. |
| $\frac{\csc x - 1}{\cot x} = \frac{\cot x}{\csc x + 1}$ | $\frac{1}{\sec x(1 - \sin x)} = \sec x + \tan x$ |
| iv. | viii. |
| $\frac{1}{\tan x + \cot x} = \cos x \sin x$ | $\frac{\tan x + \tan y}{\cot x + \cot y} = \tan y \tan x$ |

(d) Verify each of the following:

i.

$$\cos(2x) = 2\cos^2 x - 1$$

iii.

$$\cos(\theta - \pi) = -\cos \theta$$

v.

$$\sin\left(\theta + \frac{\pi}{2}\right) = \cos \theta$$

ii.

$$\cos(2x) = 1 - 2\sin^2 x$$

iv.

$$\cos\left(\theta + \frac{\pi}{2}\right) = -\sin \theta$$

vi.

$$\tan(\theta - \pi) = \tan \theta$$

3. (a) Without using a calculator or a table, solve each of the following equations for $0 \leq x \leq 2\pi$:

i.

$$2\sin x + \sqrt{3} = 0$$

iii.

$$2\cos^2 x - \sqrt{3}\cos x = 0$$

v.

$$2\cos^2 x + \cos x = \sin^2 x$$

ii.

$$\sec x = 2$$

iv.

$$\sin x + 2 = 3$$

vi.

$$\tan x = \cot x$$

(b) Without using a calculator or a table, solve each of the following equations for $0 \leq x \leq 2\pi$:

i.

$$2\tan x \sec x - \tan x = 0$$

iii.

$$\cos 2x + 3\sin x - 2 = 0$$

ii.

$$\sec^2 x - \sec x - 2 = 0$$

iv.

$$\sin x + \cos x = 1$$

(c) Sketch each of the following in the indicated interval:

i.

$$f(x) = 1 + \sin x, \quad 0 \leq x \leq 2\pi$$

iii.

$$f(x) = -\cos x, \quad 0 \leq x \leq \frac{2\pi}{3}$$

ii.

$$f(x) = 1 - \cos x, \quad 0 \leq x \leq 2\pi$$

iv.

$$f(x) = \frac{1}{2} \sin\left(\frac{1}{2}x\right), \quad 0 \leq x \leq 4\pi$$

(d) Find the period, amplitude and phase shift of each of the following functions, and in each case sketch 1 complete revolution for $x \geq 0$.

i.

$$f(x) = 2\sin 2x$$

iv.

vi.

ii.

$$f(x) = \cos\left(2x + \frac{\pi}{2}\right)$$

$$f(x) = -1 - \cos(x + 15^\circ)$$

$$f(x) = \frac{2}{3} \sin 2(x - \pi)$$

iii.

$$f(x) = 3\sin(x - 30^\circ)$$

v.

$$f(x) = -4\cos(3x + 15^\circ)$$

vii.

$$f(x) = 2 + 3\cos\left(2x - \frac{\pi}{2}\right)$$